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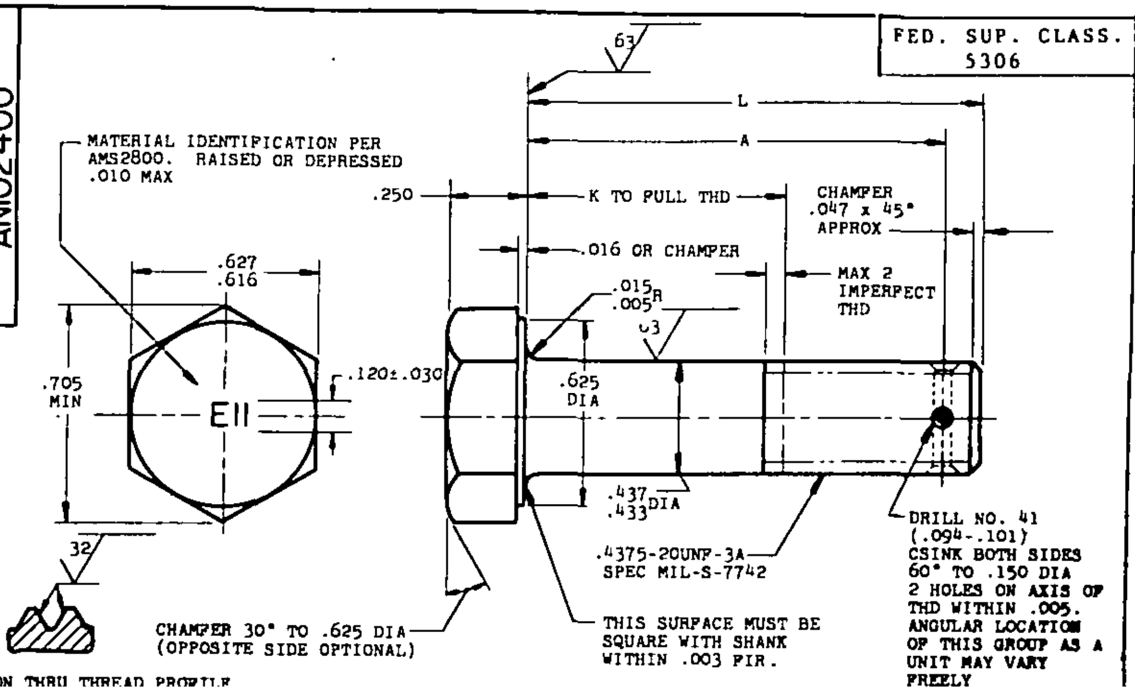
NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all antecedent standard drawings for the same product and shall become effective for the procurement of aeronautical supplies, or for use in new design, not later than 6 months after the latest date of approval shown.

AN102301
THRU
AN102400

FED. SUP. CLASS.
5306


63
L
A
MATERIAL IDENTIFICATION PER AMS2800. RAISED OR DEPRESSED .010 MAX
.250
K TO FULL THD
CHAMFER .047 x 45° APPROX
.016 OR CHAMFER
.627
.616
MAX 2 IMPERFECT THD
.015
.005
R
63
.705 MIN
E11
.120±.030
.625 DIA
.437
.433
DIA
DRILL NO. 41 (.094-.101) CSINK BOTH SIDES 60° TO .150 DIA 2 HOLES ON AXIS OF THD WITHIN .005. ANGULAR LOCATION OF THIS GROUP AS A UNIT MAY VARY FREELY
.4375-20UNF-3A SPEC MIL-S-7742
32
CHAMFER 30° TO .625 DIA (OPPOSITE SIDE OPTIONAL)
THIS SURFACE MUST BE SQUARE WITH SHANK WITHIN .003 FIR.

SECTION THRU THREAD PROFILE

L	A	K +.000 -.060	PART NO.	L	A	K +.000 -.060	PART NO.	L	A	K +.000 -.060	PART NO.
				2.125	1.985	1.312	AN102324	4.750	4.610	3.938	AN102345
				2.250	2.110	1.438	AN102325	4.875	4.735	4.062	AN102346
				2.375	2.235	1.562	AN102326	5.000	4.860	4.188	AN102347
.812	.672	(a)	AN102306	2.500	2.360	1.688	AN102327	5.125	4.985	4.312	AN102348
.875	.735	(a)	AN102307	2.625	2.485	1.812	AN102328	5.250	5.110	4.438	AN102349
.938	.798	.125	AN102308	2.750	2.610	1.938	AN102329	5.375	5.235	4.562	AN102350
1.000	.860	.188	AN102309	2.875	2.735	2.062	AN102330	5.500	5.360	4.688	AN102351
1.062	.922	.250	AN102310	3.000	2.860	2.188	AN102331	5.625	5.485	4.812	AN102352
1.125	.985	.312	AN102311	3.125	2.985	2.312	AN102332	5.750	5.610	4.938	AN102353
1.188	1.048	.375	AN102312	3.250	3.110	2.438	AN102333	5.875	5.735	5.062	AN102354
1.250	1.110	.438	AN102313	3.375	3.235	2.562	AN102334	6.000	5.860	5.188	AN102355
1.312	1.172	.500	AN102314	3.500	3.360	2.688	AN102335				
1.375	1.235	.562	AN102315	3.625	3.485	2.812	AN102336				
1.438	1.298	.625	AN102316	3.750	3.610	2.938	AN102337				
1.500	1.360	.688	AN102317	3.875	3.735	3.062	AN102338				
1.562	1.422	.750	AN102318	4.000	3.860	3.188	AN102339				
1.625	1.485	.812	AN102319	4.125	3.985	3.312	AN102340				
1.688	1.548	.875	AN102320	4.250	4.110	3.438	AN102341				
1.750	1.610	.938	AN102321	4.375	4.235	3.562	AN102342				
1.875	1.735	1.062	AN102322	4.500	4.360	3.688	AN102343				
2.000	1.860	1.188	AN102323	4.625	4.485	3.812	AN102344				

(a) THREAD TO HEAD. MAXIMUM TWO IMPERFECT THREADS.

NOTE:
(1) SHANK SHALL BE STRAIGHT WITHIN .0025 FIR. PER INCH OF BOLT LENGTH.
(2) THE CONCENTRICITY OF THREAD PD IN RELATION TO THE SHANK SHALL BE WITHIN .006 FIR.
(3) THE CONCENTRICITY OF THE SHANK IN RELATION TO THE WASHER FACE DIAMETER AND HEXAGON SHALL BE WITHIN .019 FIR.

MATERIAL: STEEL AMS6322
HARDNESS: ROCKWELL C26-32
FINISH: CADMIUM PLATE AMS2400
SURFACE ROUGHNESS: AS107
MANUFACTURING SPECIFICATION: AMS7452
③ INSPECTION: ALL PARTS SHALL UNDERGO MAGNETIC INSPECTION IN ACCORDANCE WITH AMS2640
BREAK SHARP EDGES .003-.015 UNLESS OTHERWISE SPECIFIED.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED: TOLERANCES; LINEAR DIMENSIONS ±.010, ANGULAR DIMENSIONS ±2°.
DO NOT USE UNASSIGNED PART NUMBERS.

INACTIVE FOR NEW DESIGN AFTER 3 MARCH 1969 NO SUPERSEDING STANDARD.

THIS STANDARD WAS DEVELOPED COOPERATIVELY BY THE ENGINEERING AND PROPELLER UTILITY PARTS COMMITTEE OF THE SAE

APPROVED 8 June 49 REVISED ① 31 Aug 54 ② 3 Mar 69 ③ 30 June 86

P.A. AF-11 **CUST. NAVY-AS** **ARMY-AV**	AIR FORCE-NAVY AERONAUTICAL STANDARD BOLT - HEX HEAD, DRILLED SHANK, .4375-20	AN102301 THRU AN102400

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Project No.5306-1008